An Exact BCOV Algebra in Yamaguchi–Yau Variables

Definitions, conventions, signs, normalization factors, and implementation audit

Companion to `bcov_yy.py`, version 0.2.0 (frame audit)

19 October 2026

Abstract

This note specifies the exact mathematical problem solved by the accompanying quintic BCOV implementation. It derives the differential ring in u, v_1, v_2, v_3, X, the normalized covariant derivative, and the three polynomial anomaly equations, accounting explicitly for the minus sign and factor $1/2$ in the recursion. The genus-two calculation is expanded coefficient by coefficient. Holomorphic inputs, algebraic consequences, and conventions for integration constants are distinguished throughout. The note also documents the sparse exact representation, shared arithmetic recipes, validation, and limitations of the implementation. It does not claim that unfixed boundary data have been determined by the anomaly equation.

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1 Convention ledger and exact cross-frame audit (19 October 2026)

This section records the normalization and gauge reconciliation among original Yamaguchi–Yau (YY), Huang–Klemm–Quackenbush (HKQ), and the small-radius coordinate used in the Gu/Klemm free-energy source files. The original YY BCOV recursion below remains our *canonical computational convention*. Coordinates and gauges are metadata, not implicitly interchangeable symbols.

1.1 Coordinates, Yukawa tensors, and normalized amplitudes

Write s for the original YY fifth-root coordinate, $\psi = s^5$ for HKQ, and $z = (3125\psi)^{-1}$ for the small large-radius coordinate. In particular,

$$\theta_s = 5\theta_\psi = -5\theta_z, \qquad X_{\mathrm{YY}} = X_{\mathrm{HKQ}} = \frac{1}{1-\psi}, \qquad X_z = \frac{1}{1-3125z}, \qquad X_{\mathrm{HKQ}} = 1 - X_z. \tag{1.1}$$

The holomorphic Yukawa coupling C_{ijk} is a cubic tensor: it transforms with *three* coordinate Jacobians. Set $C_t = t^3 C_{ttt}$ for each coordinate t, using a *common* choice of holomorphic three-form. Then

$$C_s = 125 C_\psi, \qquad C_\psi = -C_z, \qquad C_z = 5X_z, \quad C_{zzz} = 5X_z/z^3, \quad P_g^{(t)} = C_t^{g-1} F_g^{(t)}. \tag{1.2}$$

A further constant rescaling of the holomorphic three-form changes the individual Yukawa and section-valued free energy but cancels in the normalized P_g, provided it is implemented consistently. The $\kappa = 5$ A-model frame and factors of $(2\pi i)^3$ should not be silently combined.

Under the metric pullback $G_{\psi\bar{\psi}} = |dz/d\psi|^2 G_{z\bar{z}}$ (with the same Kähler frame),

$$A_\psi = -A_z - 2, \quad B_{p,\psi} = (-1)^p B_{p,z}, \quad A_s = 5A_\psi + 4, \quad B_{p,s} = 5^p B_{p,\psi}. \tag{1.3}$$

The polynomial generators consequently satisfy

$$(u, v_1, v_2, v_3, X)_\psi = (-u, -v_1, v_2, -v_3, 1 - X)_z, \tag{1.4}$$

$$(u, v_1, v_2, v_3, X)_s = (5u_\psi, 5v_{1,\psi}, 25v_{2,\psi}, 125v_{3,\psi}, X_\psi). \tag{1.5}$$

If T acts by substituting the variables in the right-hand side, the corresponding free-energy polynomial transformations are

$$P_g^{\mathrm{HKQ}}(v_1, v_2, v_3, X) = (-1)^{g-1} P_g^z(-v_1, v_2, -v_3, 1 - X), \tag{1.6}$$

$$P_g^{\mathrm{YY}}(v_1, v_2, v_3, X) = 5^{3g-3} P_g^{\mathrm{HKQ}}(v_1/5, v_2/25, v_3/125, X). \tag{1.7}$$

These are involutive/rational polynomial frame maps, checked symbolically at $g = 2$ and with multiple independent polynomial round trips. They include the overall Yukawa sign; changing only X is insufficient.

1.2 Propagator gauge: supplied five holomorphic functions

Set $\mu = 1/3125$, $\kappa = 5$ and use X_z in the following equations. The holomorphic functions are precisely the supplied Gu/Klemm choice:

$$q_{zz}^z = -\frac{8}{5z}, \quad q_z^{zz} = \frac{z}{25}, \quad q_z^z = -\frac{6}{625}, \quad q_{zz} = \frac{2}{25z^2}, \quad q_z = \frac{2}{3125z}. \tag{1.8}$$

In this gauge the three modified propagators are

$$S^{zz} = -\frac{z^2}{5X_z}\left(v_{1,z} + \frac{3}{5}\right), \tag{1.9}$$

$$\widetilde{S}^z = \frac{z}{5X_z}\left(v_{2,z} + \frac{2}{25}\right), \tag{1.10}$$

$$\widetilde{S} = \frac{1}{10X_z}\left[v_{3,z} + (1-X_z)\left(v_{2,z} + \frac{4}{125}\right)\right]. \tag{1.11}$$

The two derivative identities determining $\widetilde{S}^z, \widetilde{S}$ have been independently reduced to zero using exact rational SymPy expressions. The fifth q_z enters the larger BCOV differential system; it is *recorded but not tested* by those two propagator identities. The original background notes had a repeated $\widetilde{S}^z$ label on the left-hand side of the third propagator equation. It should read $\widetilde{S}$. Likewise $z\partial_z = X(X-1)\partial_X$ only *when acting on functions of X alone*.

1.3 Genus-two anchor and period/mirror-map normalization

The exact polynomial from the source `ff_2.txt` after the above coordinate transformation is the HKQ expression

$$\begin{aligned} P_2^{\mathrm{HKQ}} = {} & \frac{25}{144} - \frac{625}{288}v_1 + \frac{25}{24}v_1^2 - \frac{5}{24}v_1^3 - \frac{625}{36}v_2 + \frac{25}{6}v_1v_2 + \frac{350}{9}v_3 \\ & - \frac{5759}{3600}X - \frac{167}{720}v_1X + \frac{1}{6}v_1^2X - \frac{475}{12}v_2X + \frac{41}{3600}X^2 - \frac{13}{288}v_1X^2 + \frac{X^3}{240}. \end{aligned} \tag{1.12}$$

Substitution into the original YY frame reproduces the published P_2 and the program's `reference_genus2()`. For the Frobenius periods with $z \to 0$,

$$\omega_0(z) = \sum_{n\geq 0} \frac{(5n)!}{(n!)^5} z^n, \tag{1.13}$$

$$\sigma_1(z) = \sum_{n\geq 1} \frac{(5n)!}{(n!)^5} 5(H_{5n} - H_n) z^n, \tag{1.14}$$

$$t = \frac{\omega_1}{\omega_0}, \quad \omega_1 = \frac{\omega_0 \log z + \sigma_1}{2\pi i}, \quad q = e^{2\pi i t} = z\exp(\sigma_1/\omega_0). \tag{1.15}$$

In particular $z(q) = q - 770q^2 + 171525q^3 - 81623000q^4 + \cdots$. The correctly normalized A-model limit is

$$F_g^A = \omega_0^{2g-2}\left(\frac{1-\psi}{5\psi}\right)^{g-1} P_g^{\mathrm{HKQ}}\big|_{\mathrm{hol}}, \qquad \psi = (3125z)^{-1}. \tag{1.16}$$

Our older Mathematica `BCOVpropagators.txt` helper used a fifth-root variable in its C_{fff} definition but substituted the HKQ variable $1/(3125z)$ directly for it. That substitution omits a fifth root and, for C_{sss}, a cubic Jacobian. Its object called `Cf[z]` was *an inverse Yukawa* in that helper, whereas the source `ff_g.txt` uses $C_{zzz} = 5X_z/z^3$. These two quantities must never share an unqualified symbol in executable code.

1.4 Exact enumerative check: genera 2–15, degrees 1–6

For $g \geq 2$ the instanton contribution is extracted by

$$[q^d]F_g^A = \sum_{m|d} m^{2g-3} \sum_{h=0}^{g} n_{d/m}^h \, [y^{g-h}] \left(\frac{2\sin(\sqrt{y}/2)}{\sqrt{y}}\right)^{2h-2}. \tag{1.17}$$

For $h = 0$ the displayed coefficient prescription is $[y^g](2\sin(\sqrt{y}/2)/\sqrt{y})^{-2}$; it accounts for the λ^{-2} genus-zero contribution correctly. Genus-one GV data contribute no higher-genus coefficient because the sine power is zero. The source is `quinticGVDTPT.m` and its `GVTab`, not a manually typed subset. Running `verify_physical_qseries.py` confirms **all 84 positive-degree coefficients**, $g = 2, \ldots, 15$ and $d = 1, \ldots, 6$, with exact rational zero residuals. For example, $[q]F_2^A = 2875/240 = 575/48$: the free-energy coefficients are rational genus-g Gromov–Witten quantities, not the integer GV invariants themselves.

The 61-by-44 `GVTab` and the partial 113-by-31 `GVTabExt` disagree at 29 numerical overlaps, beginning at $(g, d) = (26, 16)$. These conflicts have *not* been adjudicated; our free-energy sources stop at genus 15. Do not treat an unknown entry in the extended table as zero, nor mistake its sparse genus-64 row for a complete genus-64 enumeration.

1.5 Bernoulli and arithmetic invariants after frame correction

Define $P_g = P_g^{\mathrm{an}} + a_g(X)$ in the original YY frame, with $P_g^{\mathrm{an}}(0, 0, 0, X) = 0$, and $a_g(X) = \sum_n a_{g,n} X^n$. The two independently re-tested formulas are

$$b_1 = \frac{1}{6}, \quad b_g = \frac{1}{2g+1} \sum_{h=1}^{g-1} \binom{2g}{2h} b_h b_{g-h}, \qquad b_g = |B_{2g}|, \tag{1.18}$$

$$a_{g,0} = 100\, 625^{g-1} \frac{b_g b_{g-1}}{2g(2g-2)(2g-2)!}. \tag{1.19}$$

For $g \geq 3$, put

$$S_g = \sum_{h=2}^{g-2} (2h-2)(2g-2h-2) a_{h,0} a_{g-h,0}, \qquad I_g = \frac{23}{25} \frac{(-625)^{g-1} b_g}{2g(2g-2)!}, \tag{1.20}$$

and then

$$a_{g,1} = -\frac{577}{625}(g-1)a_{g,0} + (2g-4)\left(62 - \frac{24g}{5}\right) a_{g-1,0} - \frac{12}{5} S_g - I_g. \tag{1.21}$$

Genus two supplies the separate seed $a_{2,1} = -28795/144$. We extracted $[X^0], \ldots, [X^4]$ from all 14 source amplitudes without full multivariate expansion. Equations (1.18) and (1.21) pass **28 of 28 independent rational equalities** for $g = 2, \ldots, 15$.

The original YY and HKQ pure coefficients are related by $a_{g,n}^{\mathrm{YY}} = 5^{3g-3} a_{g,n}^{\mathrm{HKQ}}$, so ratios $a_{g,n}/a_{g,0}$ agree in these frames. By contrast, under $X \mapsto 1 - X$ the *individual* coefficients mix by an integral, upper-triangular binomial matrix with determinant ± 1. Therefore the least common multiple of their reduced denominators is exactly invariant:

$$\operatorname*{lcm}_n \operatorname{den}(a_{g,n}^z) = \operatorname*{lcm}_n \operatorname{den}(a_{g,n}^{\mathrm{HKQ}}). \tag{1.22}$$

Between YY and HKQ, the non-5-primary part of this denominator lattice is unchanged. The assertion follows algebraically from the integral inverse matrix, and was checked against full pure-X polynomials for genera 2–8. This protects denominator-lattice patterns, not individual face-value sequences, from the coordinate inversion. General propagator gauge transformations may mix the *primitive/ambiguity split* and must be handled separately.

1.6 Provenance, regression suite, and unsolved problems

The v0.2 archive includes the unchanged YY recursion and the additional frame maps, tests, and source-derived data. The immutable `FramedPolynomial` wrapper carries the genus and frame and rejects additions of expressions with incompatible frame metadata. The principal modules are:

`bcov_yy.py`, `quintic_frames.py`, `test_frames.py`, `test_gu_gauge.py`, `audit_coefficients.py`, `audit_denominators.py`, `verify_physical_qseries.py`.

The identity checks include P_2 in both frames, the two Gu gauge derivative equations, formal round trips, Bernoulli coefficients to genus 15, full pure-X denominator lattices to genus 8, and the 84 independent q-series checks. The source free energies and source GV table both come from the same research provenance; the comparison is algebraically independent but is not a proof of mathematical correctness of all the tabulated invariants.

We have *not* determined holomorphic ambiguities without external boundary inputs, settled the 29 conflicting high-genus integers, proved a new small-integer uniqueness principle, or certified an all-genus Bernoulli formula for the middle powers $n \sim g$. The independently established fixed-low-X jet recursions through genus 53 belong to the original YY normalization and remain valid there; the present audit independently rechecks the source free energies through genus 15.

2 Scope and the distinction between inputs and consequences

The implementation specializes to the mirror quintic and to the original ψ convention of Yamaguchi and Yau (YY) [1]. The equations below refer to that convention unless a different coordinate or gauge is explicitly introduced. The four central objects must not be conflated:

$$F_g(\psi, \bar{\psi}) : \text{the B-model amplitude in a chosen Hodge-line gauge,} \tag{2.1}$$

$$P_g = C^{g-1} F_g : \text{the normalized polynomial amplitude,} \tag{2.2}$$

$$P_g^{\mathrm{an}} : \text{the implementation's particular polynomial primitive,} \tag{2.3}$$

$$F_g^A(t) : \text{the holomorphic A-model amplitude after the frame conversion.} \tag{2.4}$$

Here $C = \psi^3 C_{\psi\psi\psi}$, not $C_{\psi\psi\psi}$ itself.

The geometric inputs are the quintic Picard–Fuchs operator, special geometry, the choice of holomorphic terms that closes the differential ring, the genus-one one-point function, and any supplied higher-genus boundary data. The program derives higher anomaly polynomials from those inputs by exact polynomial arithmetic. It neither proves the special-geometry identities from the construction of the Calabi–Yau family nor independently solves all physical boundary conditions.

In particular, the four pure-X coefficients supplied at genus two are reference boundary data from the complete YY polynomial. Their presence must not be described as a prediction of the anomaly recursion alone. The note makes the origin of each such coefficient explicit.

3 Coordinates, discriminants, and Euler derivatives

The mirror family is obtained from the quintic polynomial

$$p(x, \psi) = x_1^5 + x_2^5 + x_3^5 + x_4^5 + x_5^5 - 5\psi x_1 x_2 x_3 x_4 x_5 \tag{3.1}$$

with its usual mirror quotient. Set

$$z = (5\psi)^{-5} = \frac{1}{5^5 \psi^5}, \qquad \vartheta = \psi \partial_\psi, \qquad \theta_z = z \partial_z. \tag{3.2}$$

The factor $5^5 = 3125$ is part of the definition of z. Since

$$\partial_\psi z = -\frac{5z}{\psi}, \qquad \vartheta z = -5z, \tag{3.3}$$

the chain rule gives

$$\boxed{\vartheta = -5\theta_z.} \tag{3.4}$$

This accounts for the division by -5 in `YYRing.theta_z`. It implements $z\partial_z$, not ∂_z. To obtain ∂_z one would also need multiplication by z^{-1}, which is outside the present polynomial ring unless a localization is added.

The YY generator and the conventional discriminant are

$$X = \frac{1}{1-\psi^5}, \qquad \Delta(z) = 1 - 3125z. \tag{3.5}$$

Solving for the same X in z gives

$$X = \frac{3125z}{3125z-1} = -\frac{3125z}{\Delta}, \tag{3.6}$$

$$X - 1 = -\frac{1}{\Delta}, \qquad 1 - X = \frac{1}{\Delta}. \tag{3.7}$$

Thus a paper defining its variable as $1/\Delta$ uses $1 - X$, not this X. The sign changes under this substitution are real convention changes. At large complex structure, $z \to 0$, $\psi \to \infty$, and $X \to 0$. At the conifold, $\psi^5 = 1$ and X has a pole.

Differentiating the defining rational function directly,

$$\vartheta X = \frac{5\psi^5}{(1-\psi^5)^2} = 5\left(\frac{1}{1-\psi^5}\right)\left(\frac{\psi^5}{1-\psi^5}\right) \tag{3.8}$$

$$= 5X(X-1), \tag{3.9}$$

$$\theta_z X = -X(X-1) = X(1-X). \tag{3.10}$$

The second factor is $X - 1$, since $\psi^5/(1-\psi^5) = X - 1$.

4 Hodge gauge, metric, and connection signs

The reference holomorphic-form gauge is YY's local residue representative

$$\Omega = 5\psi\,\frac{x_5\,\mathrm{d}x_1 \wedge \mathrm{d}x_2 \wedge \mathrm{d}x_3}{\partial p/\partial x_4}. \tag{4.1}$$

An overall period normalization is additionally fixed by the Yukawa coefficient specified below. The engine stores neither the form nor the periods themselves. Fix a period vector Π and symplectic matrix Σ with the convention

$$E := e^{-K} = -i\,\bar{\Pi}^{,T}\Sigma\Pi, \qquad G := G_{\psi\bar{\psi}} = \partial_\psi \bar{\partial}_{\bar{\psi}} K. \tag{4.2}$$

The overall constant in E cancels from all B_p below. Define the standard holomorphic Christoffel component by

$$\Gamma_{\mathrm{std}} = G^{-1}\partial_\psi G. \tag{4.3}$$

YY also use a Christoffel symbol with the opposite sign, $\Gamma_{\mathrm{YY}} = -\Gamma_{\mathrm{std}}$. We avoid any ambiguity by using

$$A = \vartheta \log G = \psi\Gamma_{\mathrm{std}} = -\psi\Gamma_{\mathrm{YY}}, \qquad B = \vartheta \log E = -\psi\partial_\psi K. \tag{4.4}$$

For a section h of $(T^*)^n \otimes \mathcal{L}^m$, our covariant derivative is

$$D_\psi h = \partial_\psi h - n\Gamma_{\mathrm{std}} h + mK_\psi h. \tag{4.5}$$

There is one minus Christoffel contribution for each lower holomorphic tensor index. Since $K_\psi = -B/\psi$,

$$\psi D_\psi h = \vartheta h - nAh - mBh. \tag{4.6}$$

Equivalently, YY write $+n\Gamma_{\mathrm{YY}}$; the two formulas are identical.

Under $\Omega \mapsto f(\psi)\Omega$, one has

$$K \mapsto K - \log f - \log \bar{f}, \qquad F_g \mapsto f^{2-2g} F_g. \tag{4.7}$$

The plus sign in $+mK_\psi$ in (4.5) is required: the derivative of f^m cancels the change $-m\partial_\psi \log f$ in the connection. For F_g the Hodge power is $m = 2 - 2g$.

5 Yukawa normalization and its derivative

Write

$$C_{\psi\psi\psi} = \kappa \frac{\psi^2}{1-\psi^5}, \qquad \kappa = \frac{(2\pi i)^3}{5^3} \tag{5.1}$$

in the YY period normalization. The coefficient κ is an overall period/gauge normalization; the polynomial engine never numerically evaluates it. The scalar normalization used by the code is

$$C := \psi^3 C_{\psi\psi\psi} = \kappa \frac{\psi^5}{1-\psi^5} = \kappa(X-1). \tag{5.2}$$

Consequently,

$$\vartheta \log C_{\psi\psi\psi} = 2 + 5\frac{\psi^5}{1-\psi^5} = 5X - 3, \tag{5.3}$$

$$\vartheta \log C = 3 + (5X-3) = 5X, \qquad \vartheta C = 5XC. \tag{5.4}$$

The cancellation of the $+3$ from ψ^3 against the -3 in the tensor Yukawa derivative is essential. Omitting it changes every normalized covariant derivative.

For a coordinate-only change, keeping the same Hodge gauge,

$$C_{zzz} = \left(\frac{\mathrm{d}\psi}{\mathrm{d}z}\right)^3 C_{\psi\psi\psi} = \left(-\frac{\psi}{5z}\right)^3 \kappa \frac{\psi^2}{1-\psi^5} \tag{5.5}$$

$$= -\frac{\kappa}{125z^3}(X-1) = \frac{\kappa}{125z^3\Delta}. \tag{5.6}$$

The two minus signs, one from the cubic Jacobian and one from $X - 1 = -1/\Delta$, cancel. The familiar $5/(z^3\Delta)$ is in a differently normalized Hodge gauge; it must not be inserted here without also transforming the amplitudes.

6 The original differential ring and the geometric closure input

Define, for $p \geq 1$,

$$A_p = \frac{\vartheta^p G}{G}, \qquad B_p = \frac{\vartheta^p E}{E}, \qquad A = A_1, \quad B = B_1. \tag{6.1}$$

Thus A_2 is not A^2, and B_2 is not B^2. The quotient rule gives the exact identities

$$\vartheta A_p = A_{p+1} - AA_p, \qquad \vartheta B_p = B_{p+1} - BB_p. \tag{6.2}$$

The negative terms arise from differentiating G^{-1} or E^{-1}.

6.1 Picard–Fuchs closure of B_4

The period equation in this gauge is

$$\left[\vartheta^4 - \psi^{-5}(\vartheta-1)(\vartheta-2)(\vartheta-3)(\vartheta-4)\right]\Pi = 0. \tag{6.3}$$

The elementary symmetric sums of $1, 2, 3, 4$ are $10, 35, 50, 24$, so

$$\prod_{j=1}^{4}(\vartheta - j) = \vartheta^4 - 10\vartheta^3 + 35\vartheta^2 - 50\vartheta + 24. \tag{6.4}$$

Multiplying the period equation by ψ^5 and rearranging gives

$$(1-\psi^5)\vartheta^4\Pi = 10\vartheta^3\Pi - 35\vartheta^2\Pi + 50\vartheta\Pi - 24\Pi. \tag{6.5}$$

The sign of the constant is negative on the right-hand side. Since the antiholomorphic period factor is untouched by ϑ, the same linear identity holds for E. Dividing by E therefore yields

$$\boxed{B_4 = 10XB_3 - 35XB_2 + 50XB - 24X.} \tag{6.6}$$

As a check, using (3.4), the same operator becomes

$$\theta_z^4 - 3125z\prod_{j=1}^{4}\left(\theta_z + \frac{j}{5}\right). \tag{6.7}$$

The factor $(-5)^4 = 625$ cancels in dividing the entire equation; there is no extra sign because there are four factors.

6.2 Special-geometry closure of A_2

The chosen quintic geometric closure is YY equation (3.22):

$$\boxed{\begin{aligned} A_2 = \; & -4B_2 - 2AB - 2B + 2B^2 - 2A \\ & + 10XB + 5XA - 5X - 1. \end{aligned}} \tag{6.8}$$

For clarity, $-4B_2$ and $+2B^2$ have different meanings. The origin of the holomorphic part can be exhibited by the special-geometry identity

$$\bar\partial_{\bar\psi}\left[-4B_2 - A_2 - 2AB - 2B + 2B^2 - 2A + 10XB + 5XA\right] = 0. \tag{6.9}$$

Denoting the bracket by $h(\psi)$, solving for A_2 gives the first line and the XB, XA terms in (6.8), followed by $-h$. In the adopted YY gauge the holomorphic input is

$$h = 5X + 1. \tag{6.10}$$

This explains both signs in $-5X - 1$. It is a geometric/gauge input, not a consequence of the Picard–Fuchs equation alone. Using $+5X+1$ in (6.8) would correspond to solving for the bracket with the wrong sign.

An explicit local consistency check is useful. At large complex structure,

$$\omega_0 = 1 + 120z + O(z^2), \qquad t = -\log z - 770z + O(z^2), \tag{6.11}$$

$$B = -600z + O(z^2), \qquad B_2 = 3000z + O(z^2), \tag{6.12}$$

$$A = -1 - 3850z + O(z^2), \qquad A_2 = 1 + 26950z + O(z^2), \tag{6.13}$$

$$X = -3125z + O(z^2). \tag{6.14}$$

Here $B = \vartheta \log \omega_0$ in the holomorphic limit, and $A = \vartheta \log(\partial_\psi t)$. The coefficient 770 follows from $5(120)(H_5 - H_1) = 5(120)(77/60) = 770$. Substitution in the bracket gives

$$h = 1 + [-12000 - 26950 - 1200 + 1200 + 7700 + 15625]z + O(z^2) \tag{6.15}$$

$$= 1 - 15625z + O(z^2) = 1 + 5X + O(z^2). \tag{6.16}$$

This checks the displayed constants locally. It does not, by itself, exclude arbitrary additional holomorphic functions; the exact closure is the specified geometric input.

7 Change to u, v_1, v_2, v_3, X and the complete derivation table

The precise change of variables is

$$u = B, \tag{7.1}$$
$$v_1 = A + 1 + 2B, \tag{7.2}$$
$$v_2 = B_2 - B(A + 1 + 2B) = B_2 - uv_1, \tag{7.3}$$
$$v_3 = B_3 - B\{B(1 + A + 2B) - B_2 + 5X(1 + A + 2B) - 10X\} \tag{7.4}$$
$$= B_3 + uv_2 - 5uX(v_1 - 2). \tag{7.5}$$

Its inverse is

$$\boxed{\begin{aligned} B &= u, & A &= v_1 - 1 - 2u, \\ B_2 &= v_2 + uv_1, & B_3 &= v_3 - uv_2 + 5uX(v_1 - 2). \end{aligned}} \tag{7.6}$$

The factor 5 in the last equation is present in the implementation. The constant -2 inside $v_1 - 2$ is responsible for several terms proportional to $10X$ and $24X$ below; it cannot be dropped as an irrelevant shift.

7.1 Derivative of u

Equation (6.2) gives

$$\boxed{\vartheta u = B_2 - B^2 = v_2 + uv_1 - u^2.} \tag{7.7}$$

7.2 Derivative of v_1

First differentiate before substituting:

$$\vartheta v_1 = (A_2 - A^2) + 2(B_2 - B^2) \tag{7.8}$$
$$= -2B_2 - 2AB - 2B - 2A - A^2 + 10XB + 5XA - 5X - 1. \tag{7.9}$$

The $+2B^2$ in A_2 cancels the $-2B^2$ from $2\vartheta B$. Now substitute (7.6):

$$\begin{aligned} \vartheta v_1 = {} & -2(v_2 + uv_1) - 2u(v_1 - 1 - 2u) - 2u && (7.10)\\ & - 2(v_1 - 1 - 2u) - (v_1 - 1 - 2u)^2 && (7.11)\\ & + 10Xu + 5X(v_1 - 1 - 2u) - 5X - 1 && (7.12)\\ = {} & - v_1^2 - 2v_2 - 10X + 5Xv_1. && (7.13) \end{aligned}$$

All powers of u cancel. The $-10X$ is the sum of $-5X$ from $5XA$ and the explicit $-5X$ in A_2.

7.3 Derivative of v_2

Using $v_2 = B_2 - uv_1$ gives

$$\begin{aligned} \vartheta v_2 = {} & B_3 - uB_2 - (\vartheta u)v_1 - u\vartheta v_1 && (7.14)\\ = {} & v_3 - uv_2 + 5uX(v_1 - 2) - uv_2 - u^2 v_1 && (7.15)\\ & - v_1 v_2 - uv_1^2 + u^2 v_1 && (7.16)\\ & + uv_1^2 + 2uv_2 + 10uX - 5uXv_1 && (7.17)\\ = {} & v_3 - v_1 v_2. && (7.18) \end{aligned}$$

The uv_2 coefficients are $-1 - 1 + 2 = 0$; the $u^2 v_1$ and uv_1^2 terms cancel in pairs; and $5uX(v_1 - 2) + 10uX - 5uXv_1 = 0$.

7.4 Derivative of v_3

To display all cancellations without an opaque large expansion, put

$$L = -v_2 + 5X(v_1 - 2), \qquad B_3 = v_3 + uL. \tag{7.19}$$

From (6.6) and (6.2),

$$\vartheta B_3 = (10X - u)(v_3 + uL) - 35X(v_2 + uv_1) + 50Xu - 24X. \tag{7.20}$$

Independently, using the derivatives already established,

$$\begin{aligned}
\vartheta L &= -\vartheta v_2 + 5(\vartheta X)(v_1 - 2) + 5X\vartheta v_1 && (7.21)\\
&= -v_3 + v_1v_2 + 25X(X-1)(v_1 - 2) && (7.22)\\
&\qquad + 5X(-v_1^2 - 2v_2 - 10X + 5Xv_1) && (7.23)\\
&= -v_3 + (10X - v_1)L - 35Xv_1 + 50X. && (7.24)
\end{aligned}$$

Since $v_3 = B_3 - uL$,

$$\begin{aligned}
\vartheta v_3 &= 10Xv_3 - 35Xv_2 - 24X - v_2L && (7.25)\\
&\quad + u\{-v_3 + (10X - v_1)L - 35Xv_1 + 50X - \vartheta L\} && (7.26)\\
&= 10Xv_3 - 35Xv_2 - 24X - v_2[-v_2 + 5X(v_1 - 2)] && (7.27)\\
&= v_2^2 - 24X - 25Xv_2 - 5Xv_1v_2 + 10Xv_3. && (7.28)
\end{aligned}$$

The two u^2L terms canceled before the second line. The remaining coefficient of u vanishes by the displayed expression for ϑL. Finally, $-35Xv_2 + 10Xv_2 = -25Xv_2$.

7.5 The closed table and its meaning

Combining the results,

$$\boxed{\begin{aligned}
\vartheta u &= v_2 + uv_1 - u^2,\\
\vartheta v_1 &= -v_1^2 - 2v_2 - 10X + 5Xv_1,\\
\vartheta v_2 &= -v_1v_2 + v_3,\\
\vartheta v_3 &= v_2^2 - 24X - 25Xv_2 - 5Xv_1v_2 + 10Xv_3,\\
\vartheta X &= 5X(X-1).
\end{aligned}} \tag{7.29}$$

These are ordinary derivation identities in the commutative ring

$$\mathcal{R} = \mathbb{Q}[u, v_1, v_2, v_3, X]. \tag{7.30}$$

In particular, $\mathbb{Q}[v_1, v_2, v_3, X]$ is closed under ϑ. Denote the restriction of ϑ to this subring by δ; explicitly,

$$\begin{aligned}
\delta &= (-v_1^2 - 2v_2 - 10X + 5Xv_1)\partial_{v_1} + (v_3 - v_1v_2)\partial_{v_2} && (7.31)\\
&\quad + (v_2^2 - 24X - 25Xv_2 - 5Xv_1v_2 + 10Xv_3)\partial_{v_3} + 5X(X-1)\partial_X. && (7.32)
\end{aligned}$$

For a general element of $\mathcal{R}$,

$$\vartheta = (v_2 + uv_1 - u^2)\partial_u + \delta. \tag{7.33}$$

Assign the filtration weights

$$\mathrm{wt}(u) = \mathrm{wt}(v_1) = \mathrm{wt}(X) = 1, \quad \mathrm{wt}(v_2) = 2, \quad \mathrm{wt}(v_3) = 3. \tag{7.34}$$

Each image in (7.29) has weight at most one greater than the generator. This is a filtration, not a homogeneous grading preserved by ϑ: for example, $\vartheta X = 5X^2 - 5X$ has both weights two and one. The YY bound is $\mathrm{wt}(P_g) \le 3g-3$ and $\mathrm{wt}(P_g^{(n)}) \le 3g-3+n$ for stable vertices. Indeed, assuming the bound at lower genus, both $P_{g-1}^{(2)}$ and every product $P_h^{(1)}P_{g-h}^{(1)}$ have weight at most $3g-4$. Extracting u^k reduces that bound by k. Hence the three gradient components in (10.28) have weights at most $3g-4, 3g-5, 3g-6$, respectively: multiplication of $r_{g,2}$ by X raises its weight by one in the middle component. Integration in variables of weights $1,2,3$ then gives weight at most $3g-3$ in every case. The degree restriction on the added function of X is the additional polynomial/meromorphic ansatz of the YY construction.

8 Normalized vertices and the covariant derivative: every factor

For $g \ge 2$, define

$$P_g = P_g^{(0)} = C^{g-1}F_g, \qquad P_g^{(n)} = C^{g-1}\psi^n D_\psi^n F_g. \tag{8.1}$$

The upper parenthesized index is the number of insertions; it is not a power. The stable low-genus definitions are

$$P_0^{(n)} = C^{-1}\psi^n D_\psi^{n-3} C_{\psi\psi\psi} \ (n \ge 3), \qquad P_1^{(n)} = \psi^n D_\psi^{n-1}\partial_\psi F_1 \ (n \ge 1). \tag{8.2}$$

No zero-insertion P_0 or P_1 is needed by the algorithm. In particular, $P_0^{(3)} = 1$ does not mean that the genus-zero prepotential is one.

Let $H = P_g^{(n)}$, and let h be the corresponding unnormalized rank-n vertex, so $h = C^{1-g}\psi^{-n}H$. By the product rule and (5.4),

$$\vartheta h = C^{1-g}\psi^{-n}\{\vartheta H + [5(1-g)X - n]H\}. \tag{8.3}$$

The Hodge power is $m = 2-2g$. Equation (4.5) then gives

$$\begin{aligned} C^{g-1}\psi^{n+1}D_\psi h &= \vartheta H + [5(1-g)X - n - nA - (2-2g)B]H && (8.4)\\ &= \vartheta H + [-n(A+1) + (2g-2)B - 5(g-1)X]H && (8.5)\\ &= \vartheta H + \left[-n(v_1-2u) + (2g-2)\left(u - \frac{5}{2}X\right)\right]H. && (8.6) \end{aligned}$$

Therefore

$$\boxed{P_g^{(n+1)} = \mathcal{D}_{g,n}P_g^{(n)}, \quad \mathcal{D}_{g,n} = \vartheta - n(v_1-2u) + (2g-2)\left(u - \frac{5}{2}X\right).} \tag{8.7}$$

The ledger of contributions is

Contribution	Origin
$-n$	derivative of ψ^{-n}
$-nA$	n lower covariant tensor indices
$+(2g-2)B$	Hodge power $2-2g$ and $K_\psi = -B/\psi$
$-5(g-1)X$	derivative of C^{1-g}
$-n(A+1) = -n(v_1-2u)$	inverse generator change
$5/2$	$5(g-1) = (2g-2)(5/2)$

For two derivatives, the second operator is $\mathcal{D}_{g,1}$, not another $\mathcal{D}_{g,0}$. For example,

$$P_2^{(2)} = \mathcal{D}_{2,1}\mathcal{D}_{2,0}P_2. \tag{8.8}$$

Genus is unchanged during differentiation; tensor rank increases by one.

The genus-zero seed follows by cancellation:

$$P_0^{(3)} = C^{-1}\psi^3 C_{\psi\psi\psi} = 1. \tag{8.9}$$

As a sign/rank check,

$$P_0^{(4)} = \mathcal{D}_{0,3}1 = -3(v_1 - 2u) - 2(u - 5X/2) = 5X - 3v_1 + 4u. \tag{8.10}$$

9 Genus one: the seed and the numerical constants

The Euler characteristic parameter used by YY and by this implementation is $\chi = -200$, the original quintic's value in this mirror-symmetry convention. It must not be replaced by the Euler characteristic $+200$ of the resolved mirror while keeping all other formulas unchanged. The genus-one anomaly is

$$\bar\partial_{\bar\psi}\partial_\psi F_1 = \frac{1}{2}C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi} + \left(1 - \frac{\chi}{24}\right)G. \tag{9.1}$$

Here

$$\bar C^{\psi\psi}_{\bar\psi} = e^{2K}G^{-2}\bar C_{\bar\psi\bar\psi\bar\psi}. \tag{9.2}$$

The bar label is antiholomorphic; the raised indices come from inverse metrics, not from changing the holomorphic Yukawa normalization.

The adopted genus-one holomorphic boundary data give the local derivative

$$\partial_\psi F_1 = \frac{1}{2}\partial_\psi\left[-\log G + \frac{62}{3}K + \frac{62}{3}\log\psi - \frac{1}{6}\log(1-\psi^5)\right]. \tag{9.3}$$

Terms depending only on $\bar\psi$ have no effect on this formula. The coefficient $31/3 = 2 - \chi/24$ gives $62/3$ inside the bracket because of the prefactor $1/2$. Indeed, using $\bar\partial_{\bar\psi}\partial_\psi \log G = 2G - C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi}$, the first two terms give

$$-G + \frac{1}{2}C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi} + \frac{31}{3}G = \frac{1}{2}C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi} + \frac{28}{3}G, \tag{9.4}$$

where $28/3 = 1 - \chi/24$. The $\log\psi$ and conifold-log terms are holomorphic boundary inputs, not determined by this mixed derivative.

Since $\vartheta\log(1-\psi^5) = -5(X-1)$, multiplication of (9.3) by ψ gives

$$P_1^{(1)} = -\frac{1}{2}A - \frac{31}{3}B + \frac{31}{3} + \frac{5}{12}(X-1) \tag{9.5}$$

$$= \frac{31}{3}(1-B) + \frac{5}{12}(X-1) - \frac{1}{2}A \tag{9.6}$$

$$= \boxed{\frac{125}{12} - \frac{1}{2}v_1 - \frac{28}{3}u + \frac{5}{12}X}. \tag{9.7}$$

The constants are $31/3 - 5/12 + 1/2 = 125/12$; the u coefficient is $-31/3 + 1 = -28/3$. The $+5/12$ in the conifold term comes from $\frac{1}{2}(-\frac{1}{6})(-5)$: both minus signs are needed.

10 From the geometric BCOV equation to the polynomial anomaly

For $g \geq 2$, the BCOV equation [3] is

$$\bar\partial_{\bar\psi}F_g = \frac{1}{2}\bar C^{\psi\psi}_{\bar\psi}\left[D_\psi D_\psi F_{g-1} + \sum_{h=1}^{g-1} D_\psi F_h D_\psi F_{g-h}\right]. \tag{10.1}$$

The $1/2$ is the BCOV symmetry factor. The sum as written is ordered: h and $g-h$ both occur unless they are equal.

10.1 Normalization of each term on the right

For the first term,

$$D_\psi D_\psi F_{g-1} = C^{2-g}\psi^{-2}P^{(2)}_{g-1}. \tag{10.2}$$

For each product,

$$D_\psi F_h D_\psi F_{g-h} = C^{1-h}\psi^{-1}P^{(1)}_h\, C^{1-(g-h)}\psi^{-1}P^{(1)}_{g-h} \tag{10.3}$$

$$= C^{2-g}\psi^{-2}P^{(1)}_h P^{(1)}_{g-h}. \tag{10.4}$$

The exponents add as $(1-h)+(1-g+h)=2-g$. Since C is holomorphic, $\bar\partial_{\bar\psi}(C^{g-1}F_g) = C^{g-1}\bar\partial_{\bar\psi}F_g$. Multiplying (10.1) by C^{g-1} therefore leaves precisely $C^{g-1}C^{2-g}\psi^{-2} = C/\psi^2$:

$$\bar\partial_{\bar\psi}P_g = \frac{1}{2}\frac{C}{\psi^2}\bar C^{\psi\psi}_{\bar\psi}H_g, \qquad H_g = P^{(2)}_{g-1} + \sum_{h=1}^{g-1}P^{(1)}_h P^{(1)}_{g-h}. \tag{10.5}$$

10.2 The minus sign from special geometry

Set $a=\bar\partial_{\bar\psi}A$ and $b=\bar\partial_{\bar\psi}B$. From (4.4),

$$b = -\psi G. \tag{10.6}$$

Special geometry in our Christoffel convention gives

$$\bar\partial_{\bar\psi}\Gamma_{\rm std} = 2G - C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi}. \tag{10.7}$$

Thus

$$a+2b = \psi[2G - C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi}] - 2\psi G \tag{10.8}$$

$$= -\psi C_{\psi\psi\psi}\bar C^{\psi\psi}_{\bar\psi} = -\frac{C}{\psi^2}\bar C^{\psi\psi}_{\bar\psi}. \tag{10.9}$$

Inserting this identity into (10.5),

$$\boxed{\bar\partial_{\bar\psi}P_g = -\frac{1}{2}(a+2b)H_g.} \tag{10.10}$$

The sign is negative even though (10.1) has a positive $1/2$. It comes from the relation between the antiholomorphic Yukawa and $a+2b$.

As an optional propagator check, the adopted holomorphic propagator choice is

$$S^{\psi\psi} = \frac{2K_\psi + 2/\psi - \partial_\psi \log G}{C_{\psi\psi\psi}}, \qquad T^{\psi\psi} := \frac{C}{\psi^2}S^{\psi\psi} = 2-2B-A = 3-v_1. \tag{10.11}$$

The $2/\psi$ term is the propagator-gauge contribution corresponding to the local holomorphic choice $\mathfrak{f}=\psi$; it is not a second copy of K_ψ. Since $\bar\partial_{\bar\psi}S^{\psi\psi} = \bar C^{\psi\psi}_{\bar\psi}$, one obtains again $(C/\psi^2)\bar C = -\bar\partial_{\bar\psi}v_1 = -(a+2b)$.

10.3 Antiholomorphic changes of all generators

The quotient identities and $b=-\psi G$ give

$$\bar\partial_{\bar\psi}B_2 = \bar\partial_{\bar\psi}(\vartheta B + B^2) = \vartheta b + 2Bb = (A+1+2B)b = v_1 b, \tag{10.12}$$

$$\bar\partial_{\bar\psi}B_3 = \bar\partial_{\bar\psi}(\vartheta B_2 + BB_2) \tag{10.13}$$

$$= \{\vartheta v_1 + (A+1+B)v_1 + B_2\}b \tag{10.14}$$

$$= \{-v_2 + 5X(v_1-2)\}b. \tag{10.15}$$

In the first line $\vartheta b = (A+1)b$, since $b = -\psi G$. The second simplification uses the already derived ϑv_1. Consequently,

$$\bar\partial_{\bar\psi} u = b, \tag{10.16}$$
$$\bar\partial_{\bar\psi} v_1 = a + 2b, \tag{10.17}$$
$$\bar\partial_{\bar\psi} v_2 = \bar\partial_{\bar\psi}(B_2 - uv_1) = v_1 b - v_1 b - u(a+2b) = -u(a+2b), \tag{10.18}$$
$$\bar\partial_{\bar\psi} v_3 = \bar\partial_{\bar\psi}[B_3 + uv_2 - 5uX(v_1 - 2)] \tag{10.19}$$
$$= \{-v_2 + 5X(v_1-2)\}b + v_2 b - u^2(a+2b) \tag{10.20}$$
$$- 5X(v_1-2)b - 5uX(a+2b) \tag{10.21}$$
$$= -u(u+5X)(a+2b), \tag{10.22}$$
$$\bar\partial_{\bar\psi} X = 0. \tag{10.23}$$

Every term proportional to b in the intermediate expression for $\bar\partial_{\bar\psi} v_3$ cancels; the remaining two terms factor as shown.

10.4 The two formal equations and the three gradient components

By the chain rule,

$$\bar\partial_{\bar\psi} P_g = b\partial_u P_g + (a+2b)\mathscr{L} P_g, \quad \mathscr{L} = \partial_{v_1} - u\partial_{v_2} - u(u+5X)\partial_{v_3}. \tag{10.24}$$

Comparing this with (10.10) in the formal YY anomaly system gives

$$\boxed{\partial_u P_g = 0, \qquad \mathscr{L} P_g = -\frac{1}{2} H_g.} \tag{10.25}$$

More explicitly, the coefficient of a gives $\mathscr{L} P_g = -H_g/2$; the coefficient of b gives $\partial_u P_g + 2\mathscr{L} P_g = -H_g$; subtracting twice the first equation gives $\partial_u P_g = 0$.

The coefficient comparison is a formal polynomial-ring statement used by YY. It should not be misread as claiming that two $(0,1)$ differential forms on a one-complex-dimensional base are linearly independent over all smooth functions. The implementation adopts the resulting formal differential/anomaly algebra.

Define the source used by the code by

$$R_g = -\frac{1}{2} H_g = r_{g,0} + u r_{g,1} + u^2 r_{g,2}. \tag{10.26}$$

The absence of higher powers follows from the insertion formulas discussed below and is checked by the program. Since P_g is independent of u,

$$\mathscr{L} P_g = \partial_{v_1} P_g + u[-\partial_{v_2} P_g - 5X\partial_{v_3} P_g] - u^2 \partial_{v_3} P_g. \tag{10.27}$$

Therefore

$$\boxed{\begin{aligned} \partial_{v_1} P_g &= r_{g,0}, \\ \partial_{v_2} P_g &= -r_{g,1} + 5X r_{g,2}, \\ \partial_{v_3} P_g &= -r_{g,2}. \end{aligned}} \tag{10.28}$$

In the second equation the plus sign comes from substituting $\partial_{v_3} P_g = -r_{g,2}$ into $-\partial_{v_2} P_g - 5X\partial_{v_3} P_g = r_{g,1}$. Integrating only $r_{g,0}$ would incorrectly leave arbitrary functions of v_2, v_3, X, whereas the full system leaves only a function of X.

11 Exact gradient integration and the ambiguity convention

Write $G_i = \partial_{v_i} P_g$ as obtained from (10.28). The consistency conditions are

$$\partial_{v_j} G_i = \partial_{v_i} G_j \qquad (1 \le i, j \le 3). \tag{11.1}$$

They are tested as exact polynomial equalities. A particular primitive is constructed in the order v_3, v_2, v_1:

$$Q_3 = \int G_3 \, \mathrm{d}v_3, \tag{11.2}$$

$$Q_{23} = Q_3 + \int (G_2 - \partial_{v_2} Q_3) \, \mathrm{d}v_2, \tag{11.3}$$

$$P_g^{\mathrm{an}} = Q_{23} + \int (G_1 - \partial_{v_1} Q_{23}) \, \mathrm{d}v_1. \tag{11.4}$$

Every primitive here has zero integration constant in its integration variable. The mixed-partial identities ensure that each residual has no dependence on variables already integrated; a final check verifies all three derivatives. Equivalently, the selected primitive satisfies

$$P_g^{\mathrm{an}}(v_1 = v_2 = v_3 = 0, X) = 0. \tag{11.5}$$

This is an algebraic choice of representative, not a physical boundary condition.

The complete polynomial is stored as

$$P_g = P_g^{\mathrm{an}} + a_g(X), \qquad a_g(X) = \sum_{j=0}^{3g-3} a_{g,j} X^j. \tag{11.6}$$

The vector has up to $3g-2$ entries because both endpoint powers, zero and $3g-3$, are included. This is the full polynomial kernel allowed by the coarse weight bound; geometric regularity and pole restrictions may reduce the physical ambiguity subspace.

The conventional holomorphic ambiguity in F_g obeys

$$\Delta F_g = C^{1-g} \Delta a_g(X). \tag{11.7}$$

Thus a vector of coefficients of a_g is not directly a vector of coefficients of $f_g(\psi)$ in a different meromorphic basis. Moreover, our P_g^{an} need not equal C^{g-1} times a particular Feynman-diagram representative. The reference pure-X vector includes whichever shift is required by our zero-at-$v = 0$ primitive convention.

A more restrictive meromorphic ansatz used in YY for the holomorphic addition to a Feynman-diagram representative is

$$f_g(\psi) = \sum_{j=1}^{\lfloor (2g-2)/5 \rfloor} b_{g,j} \psi^{-5j} + \sum_{j=0}^{2g-2} d_{g,j} (1 - \psi^5)^{-j}. \tag{11.8}$$

Here the symbols $d_{g,j}$ avoid confusing this basis with $a_{g,j}$. Because $\psi^{-5} = X/(X-1)$, multiplication by C^{g-1} gives

$$C^{g-1} f_g = \kappa^{g-1} \Bigg[\sum_{j=1}^{\lfloor (2g-2)/5 \rfloor} b_{g,j} X^j (X-1)^{g-1-j} \tag{11.9}$$

$$+ \sum_{j=0}^{2g-2} d_{g,j} (X-1)^{g-1} X^j \Bigg]. \tag{11.10}$$

All exponents of $X-1$ in the first sum are nonnegative for $g \geq 2$. The second sum has maximal degree $(g-1)+(2g-2)=3g-3$, explaining the coarse polynomial degree used by the code. Physical constraints can therefore select an affine subspace of the full coefficient vector. Because our primitive differs from the chosen Feynman representative by a pure-X polynomial, the displayed restrictions apply directly to differences of allowed amplitudes or to a consistently converted ambiguity, not blindly to the raw vector returned by our zero-at-$v=0$ convention.

In the API, `ambiguity=None` means unresolved. It is distinct from `ambiguity=()`, which explicitly assigns zero to every coefficient. Continuing to higher genus requires an explicit choice for every lower genus. The current code propagates rational choices, not symbolic unknowns. Across several genera the dependence on earlier ambiguity parameters is generally nonlinear, so a symbolic extension would need a suitable coefficient algebra, not merely an unresolved linear vector attached to the output.

12 The genus-two recursion, fully expanded

Put

$$p := P_1^{(1)} = \frac{125}{12} - \frac{1}{2}v_1 - \frac{28}{3}u + \frac{5}{12}X. \tag{12.1}$$

The genus-two source is

$$H_2 = P_1^{(2)} + p^2, \qquad P_1^{(2)} = \mathcal{D}_{1,1}p = \vartheta p - (v_1 - 2u)p. \tag{12.2}$$

There is one product term because the ordered sum has only $h=1$; there is no extra factor two at genus two.

12.1 Differentiate the seed and apply the rank-one connection

Using the closure table gives

$$\vartheta p = \frac{1}{2}v_1^2 - \frac{25}{3}v_2 + \frac{35}{12}X - \frac{5}{2}v_1X + \frac{25}{12}X^2 - \frac{28}{3}uv_1 + \frac{28}{3}u^2. \tag{12.3}$$

For example, the v_2 coefficient is $1-28/3=-25/3$; the X coefficient is $5-25/12=35/12$. The connection contribution is

$$-(v_1-2u)p = -\frac{125}{12}v_1 + \frac{1}{2}v_1^2 - \frac{5}{12}v_1X + \frac{125}{6}u \tag{12.4}$$

$$+\left(\frac{28}{3}-1\right)uv_1 - \frac{56}{3}u^2 + \frac{5}{6}uX. \tag{12.5}$$

Adding yields

$$P_1^{(2)} = v_1^2 - \frac{25}{3}v_2 - \frac{125}{12}v_1 + \frac{35}{12}X - \frac{35}{12}v_1X + \frac{25}{12}X^2 \tag{12.6}$$

$$+u\left(\frac{125}{6} - v_1 + \frac{5}{6}X\right) - \frac{28}{3}u^2. \tag{12.7}$$

In particular the uv_1 coefficient is $-28/3+28/3-1=-1$.

12.2 Square the one-point seed

The square is an ordinary commutative polynomial square:

$$p^2 = \frac{15625}{144} - \frac{125}{12}v_1 + \frac{1}{4}v_1^2 + \frac{625}{72}X - \frac{5}{12}v_1X + \frac{25}{144}X^2 \tag{12.8}$$

$$+u\left(-\frac{1750}{9} + \frac{28}{3}v_1 - \frac{70}{9}X\right) + \frac{784}{9}u^2. \tag{12.9}$$

Every cross term carries its factor two. For example, $2(125/12)(5/12) = 625/72$, and $2(-1/2)(-28/3) = 28/3$.

12.3 Collect the source and apply the negative half

Adding (12.7) to the square gives

$$[u^0]H_2 = \frac{15625}{144} - \frac{125}{6}v_1 + \frac{5}{4}v_1^2 - \frac{25}{3}v_2 + \frac{835}{72}X - \frac{10}{3}v_1X + \frac{325}{144}X^2, \tag{12.10}$$

$$[u^1]H_2 = -\frac{3125}{18} + \frac{25}{3}v_1 - \frac{125}{18}X, \tag{12.11}$$

$$[u^2]H_2 = \frac{700}{9}. \tag{12.12}$$

For example,

$$\frac{35}{12} + \frac{625}{72} = \frac{835}{72}, \qquad \frac{25}{12} + \frac{25}{144} = \frac{325}{144}, \tag{12.13}$$

$$\frac{125}{6} - \frac{1750}{9} = -\frac{3125}{18}, \qquad -\frac{28}{3} + \frac{784}{9} = \frac{700}{9}. \tag{12.14}$$

Since $R_2 = -H_2/2$,

$$r_{2,0} = -\frac{15625}{288} + \frac{125}{12}v_1 - \frac{5}{8}v_1^2 + \frac{25}{6}v_2 - \frac{835}{144}X + \frac{5}{3}v_1X - \frac{325}{288}X^2, \tag{12.15}$$

$$r_{2,1} = \frac{3125}{36} - \frac{25}{6}v_1 + \frac{125}{36}X, \tag{12.16}$$

$$r_{2,2} = -\frac{350}{9}. \tag{12.17}$$

Applying (10.28) now gives

$$\partial_{v_1}P_2 = r_{2,0}, \tag{12.18}$$

$$\partial_{v_2}P_2 = -\frac{3125}{36} + \frac{25}{6}v_1 - \frac{2375}{12}X, \tag{12.19}$$

$$\partial_{v_3}P_2 = \frac{350}{9}. \tag{12.20}$$

The X coefficient in the second line is

$$-\frac{125}{36} + 5\left(-\frac{350}{9}\right) = -\frac{125 + 7000}{36} = -\frac{2375}{12}. \tag{12.21}$$

The positive $350/9$ in the third line is the second minus sign: $\partial_{v_3}P_2 = -r_{2,2}$.

12.4 Integrate and list all ten anomaly coefficients

First integrate G_3, then G_2, then the remaining part of G_1:

$$Q_3 = \frac{350}{9}v_3, \tag{12.22}$$

$$Q_{23} = Q_3 - \frac{3125}{36}v_2 + \frac{25}{6}v_1v_2 - \frac{2375}{12}v_2X, \tag{12.23}$$

$$P_2^{\text{an}} = Q_{23} - \frac{15625}{288}v_1 + \frac{125}{24}v_1^2 - \frac{5}{24}v_1^3 - \frac{835}{144}v_1X \tag{12.24}$$

$$+ \frac{5}{6}v_1^2X - \frac{325}{288}v_1X^2. \tag{12.25}$$

For the last line, the quadratic term of G_1 integrates as $\int(-5v_1^2/8)\mathrm{d}v_1 = -5v_1^3/24$; the mixed term integrates as $\int(5v_1X/3)\mathrm{d}v_1 = 5v_1^2X/6$. The $25v_2/6$ part of G_1 has already been supplied by $\partial_{v_1}(25v_1v_2/6)$ and must not be integrated a second time.

Monomial	Coefficient	Direct source in gradient integration
v_1	$-15625/288$	constant term of G_1
v_1^2	$125/24$	$(125/12)/2$
v_1^3	$-5/24$	$(-5/8)/3$
v_2	$-3125/36$	constant term of G_2
v_1v_2	$25/6$	v_1 term of G_2
v_3	$350/9$	G_3
v_1X	$-835/144$	X term of G_1
v_1^2X	$5/6$	$(5/3)/2$
v_2X	$-2375/12$	X term of G_2
v_1X^2	$-325/288$	X^2 term of G_1

12.5 The four supplied pure-X coefficients

The reference choice is

$$a_2(X) = \frac{3125}{144} - \frac{28795}{144}X + \frac{205}{144}X^2 + \frac{25}{48}X^3. \tag{12.26}$$

These are, in order, the four entries of `YY_GENUS2_AMBIGUITY`. The full polynomial is

$$\boxed{\begin{aligned} P_2 = {} & \frac{3125}{144} - \frac{15625}{288}v_1 + \frac{125}{24}v_1^2 - \frac{5}{24}v_1^3 - \frac{3125}{36}v_2 + \frac{25}{6}v_1v_2 + \frac{350}{9}v_3 \\ & - \frac{28795}{144}X - \frac{835}{144}v_1X + \frac{5}{6}v_1^2X - \frac{2375}{12}v_2X + \frac{205}{144}X^2 - \frac{325}{288}v_1X^2 + \frac{25}{48}X^3. \end{aligned}} \tag{12.27}$$

Equation (12.27) agrees with YY equation (4.1). The anomaly equations annihilate every term in (12.26). Consequently they cannot determine $-28795/144$, $205/144$, or $25/48$, or the constant $3125/144$, without further information. The program imports the complete reference vector rather than claiming a boundary-fitting derivation of those three remaining numbers.

There is a useful independent check of the constant coefficient. In the A-model convention specified in Section 15, the genus-two constant-map value is $F_2^A(0) = \chi/5760 = -5/144$. The conversion factor at large complex structure is $-1/625$, so

$$-\frac{1}{625}\,a_{2,0} = -\frac{5}{144} \quad \Longrightarrow \quad a_{2,0} = \frac{3125}{144}. \tag{12.28}$$

This verifies the imported constant within the stated A-model sign convention. It is not an implementation of all genus-two boundary constraints.

13 Why the source is quadratic in u and an all-genus check

For $g \geq 2$ let $k = 2g-2$, $p = P_g$, and define

$$\mathcal{Q}_g = \delta - \frac{5}{2}kX = \delta - 5(g-1)X. \tag{13.1}$$

Here p is independent of u, so

$$P_g^{(1)} = \mathcal{Q}_g p + kup. \tag{13.2}$$

Applying $\mathcal{D}_{g,1}$ and using $\vartheta u = v_2 + uv_1 - u^2$ gives

$$\boxed{\begin{aligned} P_g^{(2)} = {} & (\delta - v_1 - \tfrac{5}{2}kX)\mathcal{Q}_g p + kv_2 p \\ & + (2k+2)u\,\mathcal{Q}_g p + k(k+1)u^2 p. \end{aligned}} \tag{13.3}$$

For completeness, the linear-u terms before simplification are

$$k\delta p - \frac{5}{2}k^2 Xp + (k+2)\mathcal{Q}_g p = (2k+2)\mathcal{Q}_g p, \tag{13.4}$$

and the quadratic coefficient is $-k + k(k+2) = k(k+1)$. The $+kuv_1p$ from $k(\vartheta u)p$ cancels the $-kuv_1p$ from the rank-one connection. The genus-one seed is separately linear in u, and (12.7) is quadratic. Therefore every term of H_g is at most quadratic in u.

For an entirely explicit higher-genus coefficient prescription, set $s = 125/12 - v_1/2 + 5X/12$, $c = -28/3$, and, for $h \geq 2$,

$$k_h = 2h-2, \qquad q_h = \mathcal{Q}_h P_h, \qquad t_h = (\delta - v_1 - \tfrac{5}{2}k_h X)q_h + k_h v_2 P_h. \tag{13.5}$$

For $g \geq 3$, with $m = g-1$, collecting the endpoint terms separately gives

$$[u^0]H_g = t_m + 2sq_m + \sum_{h=2}^{g-2} q_h q_{g-h}, \tag{13.6}$$

$$[u^1]H_g = (2k_m+2)q_m + 2(sk_m P_m + cq_m) \tag{13.7}$$

$$+ \sum_{h=2}^{g-2}(k_h P_h q_{g-h} + k_{g-h}P_{g-h}q_h), \tag{13.8}$$

$$[u^2]H_g = k_m(k_m+1)P_m + 2ck_m P_m \tag{13.9}$$

$$+ \sum_{h=2}^{g-2} k_h k_{g-h} P_h P_{g-h}. \tag{13.10}$$

Each interior sum is ordered, and it is empty at $g = 3$. The endpoint factor two is the pair $h = 1, m$ and $h = m, 1$; it is not present at genus two, where they are the same single term. Multiplying each of these three lines by $-1/2$ gives $r_{g,0}, r_{g,1}, r_{g,2}$, respectively. These formulas provide a direct route to a future blocked implementation without first building an expanded polynomial in u.

The coefficient c of u in the genus-one seed is

$$c = \frac{\chi}{24} - 1 = -\frac{28}{3}. \tag{13.11}$$

For $g = 2$, $[u^2]H_2 = c + c^2 = c(c+1)$, hence

$$\partial_{v_3} P_2 = \frac{1}{2}c(c+1) = \frac{1}{2}\left(-\frac{28}{3}\right)\left(-\frac{25}{3}\right) = \frac{350}{9}. \tag{13.12}$$

For $g \geq 3$, put $a = 2g-4$. Equation (13.3) and the two endpoint terms $h = 1, g-1$ of the ordered genus sum yield

$$[u^2]H_g = a(a+1)P_{g-1} + 2caP_{g-1} \tag{13.13}$$

$$+ \sum_{h=2}^{g-2}(2h-2)(2g-2h-2)P_h P_{g-h}. \tag{13.14}$$

As $\partial_{v_3}P_g = [u^2]H_g/2$, this becomes

$$\boxed{\begin{aligned}\partial_{v_3}P_g &= (g-2)\left(2g-5+\frac{\chi}{12}\right)P_{g-1} \\ &\quad + 2\sum_{h=2}^{g-2}(h-1)(g-h-1)P_h P_{g-h}, \qquad g \geq 3.\end{aligned}} \tag{13.15}$$

The interior sum is still ordered. Its prefactor two arises from $(2h-2)(2g-2h-2)/2$, not from reducing the ordered sum to a half-sum. For the first two cases,

$$\partial_{v_3} P_3 = -\frac{47}{3} P_2, \tag{13.16}$$

$$\partial_{v_3} P_4 = -\frac{82}{3} P_3 + 2P_2^2. \tag{13.17}$$

The coefficients follow from $\chi/12 = -50/3$: $1(1-50/3) = -47/3$, $2(3-50/3) = -82/3$. These identities are tested independently of direct residual evaluation.

They can also be derived from YY equation (4.7) for

$$Z = \exp\left(\sum_{g\geq 2} \lambda^{2g-2} P_g\right), \qquad \mathcal{E} = \lambda\partial_\lambda : \tag{13.18}$$

$$\frac{2}{\lambda^2}\partial_{v_3} Z = \left[\left(\frac{\chi}{12} - 1\right)\mathcal{E} + \mathcal{E}^2 + \frac{\chi}{24}\left(\frac{\chi}{24} - 1\right)\right] Z. \tag{13.19}$$

Writing $W = \log Z$, division by Z uses $\mathcal{E}^2 Z/Z = \mathcal{E}^2 W + (\mathcal{E}W)^2$. The constant term gives genus two, the coefficient of λ^{2g-4} gives (13.15). The formal coupling λ counts genus; it is unrelated to the modulus z and to polynomial weights.

14 Genus three and the role of supplied genus-two data

At genus three the source is

$$R_3 = -\frac{1}{2}\left[P_2^{(2)} + 2P_1^{(1)} P_2^{(1)}\right]. \tag{14.1}$$

The factor two occurs because the ordered terms $h = 1$ and $h = 2$ are equal. The genus-two polynomial here includes its chosen $a_2(X)$: although a_2 is annihilated by $\mathscr{L}$, it is not annihilated by δ or by covariant differentiation. For example,

$$\delta a_2(X) = 5X(X-1)a_2'(X). \tag{14.2}$$

Consequently genus-three anomaly coefficients depend on the genus-two boundary vector. Choosing zero genus-two ambiguity instead would produce a different, mathematically consistent formal continuation.

With (12.26), the selected P_3^{an} has 55 nonzero terms, weight at most six, no u, and no pure-X terms. It is determined up to

$$a_3(X) = a_{3,0} + a_{3,1}X + \cdots + a_{3,6}X^6. \tag{14.3}$$

The code leaves these seven entries unresolved. In particular,

$$[v_3^2]P_3^{\mathrm{an}} = \frac{1}{2}\left(-\frac{47}{3}\right)\frac{350}{9} = -\frac{8225}{27}. \tag{14.4}$$

The factor $1/2$ here is from integrating a term proportional to v_3, not an additional BCOV symmetry factor. The complete list of the 55 coefficients is given in Appendix B.

15 Holomorphic and A-model limits: factors not used by the engine

The recursion produces the normalized B-model object. Recovering the B-model amplitude in the same gauge is simply

$$F_g = C^{1-g} P_g = \kappa^{1-g}(X-1)^{1-g} P_g. \tag{15.1}$$

For the holomorphic limit based at large complex structure,

$$E^{\rm hol} \propto \omega_0, \qquad G^{\rm hol} \propto \partial_\psi t, \tag{15.2}$$

where proportionality factors are antiholomorphic or independent of ψ for the relevant logarithmic derivatives. Thus one may evaluate the generators by

$$u^{\rm hol} = \frac{\vartheta\omega_0}{\omega_0}, \tag{15.3}$$

$$v_1^{\rm hol} = \frac{\vartheta^2 t}{\vartheta t} + 2\frac{\vartheta\omega_0}{\omega_0}, \tag{15.4}$$

$$v_2^{\rm hol} = \frac{\vartheta^2\omega_0}{\omega_0} - u^{\rm hol}v_1^{\rm hol}, \tag{15.5}$$

$$v_3^{\rm hol} = \frac{\vartheta^3\omega_0}{\omega_0} + u^{\rm hol}v_2^{\rm hol} - 5u^{\rm hol}X(v_1^{\rm hol}-2). \tag{15.6}$$

The first term in $v_1^{\rm hol}$ follows from $\vartheta\log(\partial_\psi t) = \vartheta\log(\vartheta t) - 1$; its -1 cancels the $+1$ in $v_1 = A + 1 + 2B$.

In the YY A-model normalization,

$$F_g^A(t) = \left[\frac{(2\pi i)^3\omega_0^2}{5^7 C}\right]^{g-1} P_g^{\rm hol} \tag{15.7}$$

$$= \left[\frac{\omega_0^2}{5^4(X-1)}\right]^{g-1} P_g^{\rm hol} \tag{15.8}$$

$$= \left[\frac{\omega_0^2(\psi^{-5}-1)}{5^4}\right]^{g-1} P_g^{\rm hol}. \tag{15.9}$$

The simplification uses $C = (2\pi i)^3(X-1)/5^3$, leaving $5^{7-3} = 5^4$; also $1/(X-1) = \psi^{-5} - 1$. At large complex structure the factor in brackets tends to $-1/625$, explaining the sign used in the genus-two constant-map check.

Equivalently, YY define $W_1^3 = \omega_0^2/[625(X-1)]$, $V_i = W_1^i v_i^{\rm hol}$, and $Y_1 = W_1 X$. If a monomial in P_g has weight w, its contribution to F_g^A becomes that monomial in V_1, V_2, V_3, Y_1 multiplied by W_1^{3g-3-w}. This is the homogeneous five-generator A-model description; it is different from the inhomogeneous four-generator polynomial P_g.

Neither the period series, the mirror map, nor these frame factors are numerically evaluated by the prototype. They are subsequent exact-series operations. The distinction is particularly important when comparing polynomials quoted in different papers.

16 Exact sparse representation and the meaning of atomic recipes

An element is represented by

$$P = \sum_{\alpha\in\mathbb{N}^5} c_\alpha x^\alpha, \quad x = (u, v_1, v_2, v_3, X), \quad c_\alpha \in \mathbb{Q}, \tag{16.1}$$

with the exponent order fixed exactly as displayed. Only nonzero coefficients are stored. A coefficient is a reduced Python `Fraction` with integer numerator and positive denominator. A negative sign belongs to the numerator. For example,

$$-\frac{325}{288}v_1X^2 \quad\longleftrightarrow\quad (0,1,0,0,2) \mapsto \text{Fraction}(-325, 288). \tag{16.2}$$

Float input is rejected rather than converted into a rational approximation. An exponent tuple has five nonnegative integers; negative exponents and polynomial denominators are not supported by this class.

The elementary algorithms are the mathematical rules

$$(P+Q)_\alpha = p_\alpha + q_\alpha, \tag{16.3}$$

$$(PQ)_\gamma = \sum_{\alpha+\beta=\gamma} p_\alpha q_\beta, \tag{16.4}$$

$$\partial_{x_i}(c_\alpha x^\alpha) = c_\alpha \alpha_i x^{\alpha-e_i}, \tag{16.5}$$

$$\int c_\alpha x^\alpha \,\mathrm{d}x_i = \frac{c_\alpha}{\alpha_i+1} x^{\alpha+e_i}, \tag{16.6}$$

$$\vartheta(c_\alpha x^\alpha) = c_\alpha \sum_i \alpha_i x^{\alpha-e_i}\vartheta x_i. \tag{16.7}$$

The last formula has no derivative of c_α because the coefficient field is $\mathbb{Q}$. Dependence on the modulus is encoded by the generators, in particular X, rather than by an extra rational function of z in each coefficient. This is why the earlier general $\mathbb{Q}(z)$ coefficient layer is unnecessary for the normalized quintic computation.

All equal monomials are combined and zero coefficients discarded after each operation. The polynomial layer is expanded and canonical. It is not an unevaluated symbolic expression tree. Scalar division remains exact; division by a nonconstant polynomial would require a separate localized/rational-function layer.

Above it, `RecipeArena` stores a directed acyclic graph with nodes for polynomial leaves, addition, multiplication, and normalized covariant derivatives labeled by (g, n). Equal node specifications are interned; commutative binary arguments are ordered by node identifiers. Thus repeated subexpressions and derivative vertices share nodes. This is structural sharing, not general factorization or discovery of arbitrary algebraic identities. `expand()` evaluates the needed graph in topological order and caches exact polynomial values. There is no recursive descent through long graph chains during this evaluation.

The source sum exploits its symmetry by iterating $1 \le h \le \lfloor g/2 \rfloor$ with multiplier two for $h < g-h$ and multiplier one for $h = g-h$. For example,

$$H_4 = P_3^{(2)} + 2P_1^{(1)}P_3^{(1)} + (P_2^{(1)})^2. \tag{16.8}$$

The central term is not doubled. The overall factor $-1/2$ is applied after assembling this bracket.

The prototype expands the source before extracting its three u coefficients. Hence its atomic representation delays and shares computations but does not eliminate expansion. The monomial-derivative cache is bounded by 8192 entries by default. The recipe arena and expanded node values can grow until explicitly compacted; `compact_recipes()` drops that arena and its cached vertices while preserving amplitudes. Externally held old recipe handles can keep an old arena alive.

The polynomial algebra itself is commutative. If one formally adjoins the operator ϑ, then $[\vartheta, M_f] = M_{\vartheta f}$ for multiplication by f. No arbitrary noncommutative operator algebra is needed by this implementation.

17 API correspondence, exact checks, and extension boundaries

Code entry point	Mathematical operation or invariant
`Poly`	Canonical sparse element of $\mathbb{Q}[u, v_1, v_2, v_3, X]$.
`partial` / `integrate`	Polynomial derivative / zero-constant primitive in one named generator.
`coefficient("u", k)`	Extract $[u^k]P$ and remove the u^k exponent.

Code entry point	Mathematical operation or invariant
`substitute`	Simultaneous substitution; replacements do not substitute into each other.
`degree()`	Maximum filtration weight; the zero polynomial uses sentinel -1.
`YYRing.theta`	Leibniz derivation from (7.29).
`YYRing.theta_z`	$-\vartheta/5$, not ∂_z.
`YYRing.covariant(p,g,n)`	$\mathcal{D}_{g,n}p$ from (8.7).
`BCOV.vertex(g,n)`	Shared recipe for the stable normalized insertion vertex.
`source_recipe(g)`	Shared recipe for $R_g = -H_g/2$.
`anomaly_gradient(g)`	All three derivatives in (10.28).
`integrate(g)`	Mixed-partial checks and the particular primitive P_g^{an}.
`set_ambiguity(g, coeffs)`	Assign rational coefficients of $a_g(X)$; invalidate affected derivative caches.
`Amplitude.polynomial()`	Return the complete chosen P_g, or reject an unresolved vector.
`residual(g,p)`	$\mathscr{L}p - R_g$; exact zero is necessary for the polynomial anomaly equation.
`save` / `load`	Versioned exact JSON amplitudes; restore checks conventions and recomputes anomaly representatives.

The residual alone does not test $\partial_u p = 0$ for an arbitrary externally supplied p; the amplitude constructor and the tests impose that condition separately. Likewise a vanishing residual does not determine $a_g(X)$. JSON numerator/denominator strings preserve exact integers. Saving uses atomic file replacement; no executable pickle is used. Restoring validates the mathematical anomaly representatives against the recorded lower-genus choices, but it does not certify the physical correctness of those choices.

The 25 tests cover rational arithmetic, rejection of floats, sparse operations, the Leibniz rule, original A, B_p identities, the filtration, tensor rank, recipe sharing, gradient integrability, the full published P_2, the independent relations (13.15), ambiguity/cache behavior, and checkpoint restoration. With the reference genus-two vector and explicit zero ambiguity choices at genera three through seven, a formal run to genus eight gives anomaly-term counts

$$10,\ 55,\ 173,\ 403,\ 830,\ 1502,\ 2548 \qquad (g = 2, \dots, 8). \tag{17.1}$$

Every residual vanishes exactly. These counts describe the specified formal run, not a physical boundary-complete genus-eight calculation or a proof of performance at genus 64.

Useful extensions have clear boundaries: a faster exact coefficient backend; blocked sparse multiplication; direct use of (13.3) to avoid materializing powers of u; modular arithmetic with certified rational reconstruction; exact boundary-condition solvers; and exact period/mirror-map series evaluation. A different Calabi–Yau model requires replacing the closure table, normalization, seed, anomaly vector fields, and ambiguity space together. Changing χ alone does not generalize the quintic engine.

A Coordinate changes of the metric connection

For comparisons in the same Hodge gauge, define

$$\widetilde{A} = \theta_z \log G_{z\bar{z}}, \qquad \widetilde{B}_p = \frac{\theta_z^p E}{E}. \tag{A.1}$$

The metric transforms as $G_{\psi\bar{\psi}} = |\partial_\psi z|^2 G_{z\bar{z}}$. Only the holomorphic Jacobian contributes to $\vartheta \log G$:

$$A = -5\widetilde{A} + \vartheta \log(-5z/\psi) = -5\widetilde{A} - 6, \tag{A.2}$$

$$B_p = (-5)^p \widetilde{B}_p. \tag{A.3}$$

The -6 consists of -5 from $\vartheta \log z$ and -1 from $-\vartheta \log \psi$. Consequently,

$$u = -5\widetilde{B}_1, \tag{A.4}$$

$$v_1 = -5(\widetilde{A} + 1 + 2\widetilde{B}_1), \tag{A.5}$$

$$v_2 = 25\{\widetilde{B}_2 - \widetilde{B}_1(\widetilde{A} + 1 + 2\widetilde{B}_1)\}. \tag{A.6}$$

For v_3, insert these expressions into $B_3 + uv_2 - 5uX(v_1 - 2)$ before making any naming convention for new z-generators. In particular, the constant -2 does not scale with v_1. A Hodge-gauge change adds its own shift to B; the above formulas describe a coordinate change alone. This is why replacing ϑ by $-5\theta_z$ without transforming the metric connection is insufficient.

B Complete genus-three anomaly coefficient table

The following table is generated directly from the exact implementation with the genus-two boundary vector (12.26). Its entries specify $P_3^{\mathrm{an}} = \sum c_m m$, with no pure-$X$ terms. Each rational is exact. It is not a claim that $a_3(X) = 0$ is the physical choice.

Monomial m	Coefficient c_m	Weight
v_2	$\frac{390625}{864}$	2
$v_1 X$	$-\frac{2882375}{576}$	2
v_3	$-\frac{146875}{432}$	3
$v_2 X$	$-\frac{808975}{144}$	3
$v_1 X^2$	$-\frac{497125}{576}$	3
$v_1 v_2$	$-\frac{1990625}{1728}$	3
$v_1^2 X$	$\frac{2917175}{3456}$	3
v_1^3	$-\frac{1953125}{10368}$	3
$v_3 X$	$\frac{1353365}{432}$	4
$v_2 X^2$	$-\frac{1736425}{108}$	4
v_2^2	$-\frac{359375}{288}$	4
$v_1 X^3$	$-\frac{570875}{1728}$	4
$v_1 v_3$	$\frac{734375}{864}$	4

Monomial m	Coefficient c_m	Weight
v_1v_2X	$-\frac{6462385}{1728}$	4
$v_1^2X^2$	$\frac{1362425}{3456}$	4
$v_1^2v_2$	$-\frac{78125}{432}$	4
v_1^3X	$-\frac{801575}{3456}$	4
v_1^4	$\frac{15625}{288}$	4
v_3X^2	$-\frac{9635}{432}$	5
v_2X^3	$\frac{1525}{27}$	5
v_2v_3	$\frac{146875}{108}$	5
v_2^2X	$-\frac{2951365}{432}$	5
v_1X^4	$-\frac{200375}{1728}$	5
v_1v_3X	$\frac{39245}{432}$	5
$v_1v_2X^2$	$-\frac{460135}{1728}$	5
$v_1v_2^2$	$-\frac{26375}{216}$	5
$v_1^2X^3$	$\frac{560425}{3456}$	5
$v_1^2v_3$	$-\frac{5875}{72}$	5
$v_1^2v_2X$	$\frac{95315}{432}$	5
$v_1^3X^2$	$-\frac{394925}{3456}$	5
$v_1^3v_2$	$\frac{8125}{144}$	5
v_1^4X	$\frac{3995}{96}$	5
v_1^5	$-\frac{625}{96}$	5
v_3X^3	$-\frac{1175}{144}$	6
v_3^2	$-\frac{8225}{27}$	6
v_2X^4	$\frac{2375}{288}$	6
v_2v_3X	$\frac{111625}{36}$	6
$v_2^2X^2$	$-\frac{6774925}{864}$	6

Monomial m	Coefficient c_m	Weight
v_2^3	$-\frac{5225}{81}$	6
v_1X^5	$-\frac{11875}{288}$	6
$v_1v_3X^2$	$\frac{15275}{864}$	6
$v_1v_2X^3$	$-\frac{31775}{1728}$	6
$v_1v_2v_3$	$-\frac{1175}{18}$	6
$v_1v_2^2X$	$\frac{56905}{216}$	6
$v_1^2X^4$	$\frac{163375}{3456}$	6
$v_1^2v_3X$	$-\frac{235}{18}$	6
$v_1^2v_2X^2$	$\frac{775}{108}$	6
$v_1^2v_2^2$	$\frac{1465}{72}$	6
$v_1^3X^3$	$-\frac{317375}{10368}$	6
$v_1^3v_3$	$\frac{235}{72}$	6
$v_1^3v_2X$	$\frac{925}{144}$	6
$v_1^4X^2$	$\frac{875}{72}$	6
$v_1^4v_2$	$-\frac{15}{4}$	6
v_1^5X	$-\frac{275}{96}$	6
v_1^6	$\frac{5}{16}$	6

C Reproducibility and source identification

The accompanying audit script `verify_bcov_notes.py` checks the expanded genus-one second vertex, the three genus-two source coefficients, the complete genus-two reference polynomial, the general quadratic-insertion identity, the genus-three/four v_3 relations, and the frame-factor simplifications. The SHA-256 digest below identifies the original 0.1.0 core implementation that was audited on 16 October 2026. The additional frame and enumerative checks in Section 1 use separate v0.2 modules:

`bd9d0415c8cc0a0fef5b8dd8e085f41538d6115cdb978f3ca569dccb86768313`

The formulas are based on YY equations (2.4), (2.12), (2.16), (2.19)–(2.20), (3.1)–(3.8), (3.16), (3.22)–(3.30), (3.31)–(3.33), (4.1), and (4.7). The BCOV equation is geometric input; the coefficient arithmetic and the implementation-specific primitive/ambiguity split are given explicitly here.

References

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